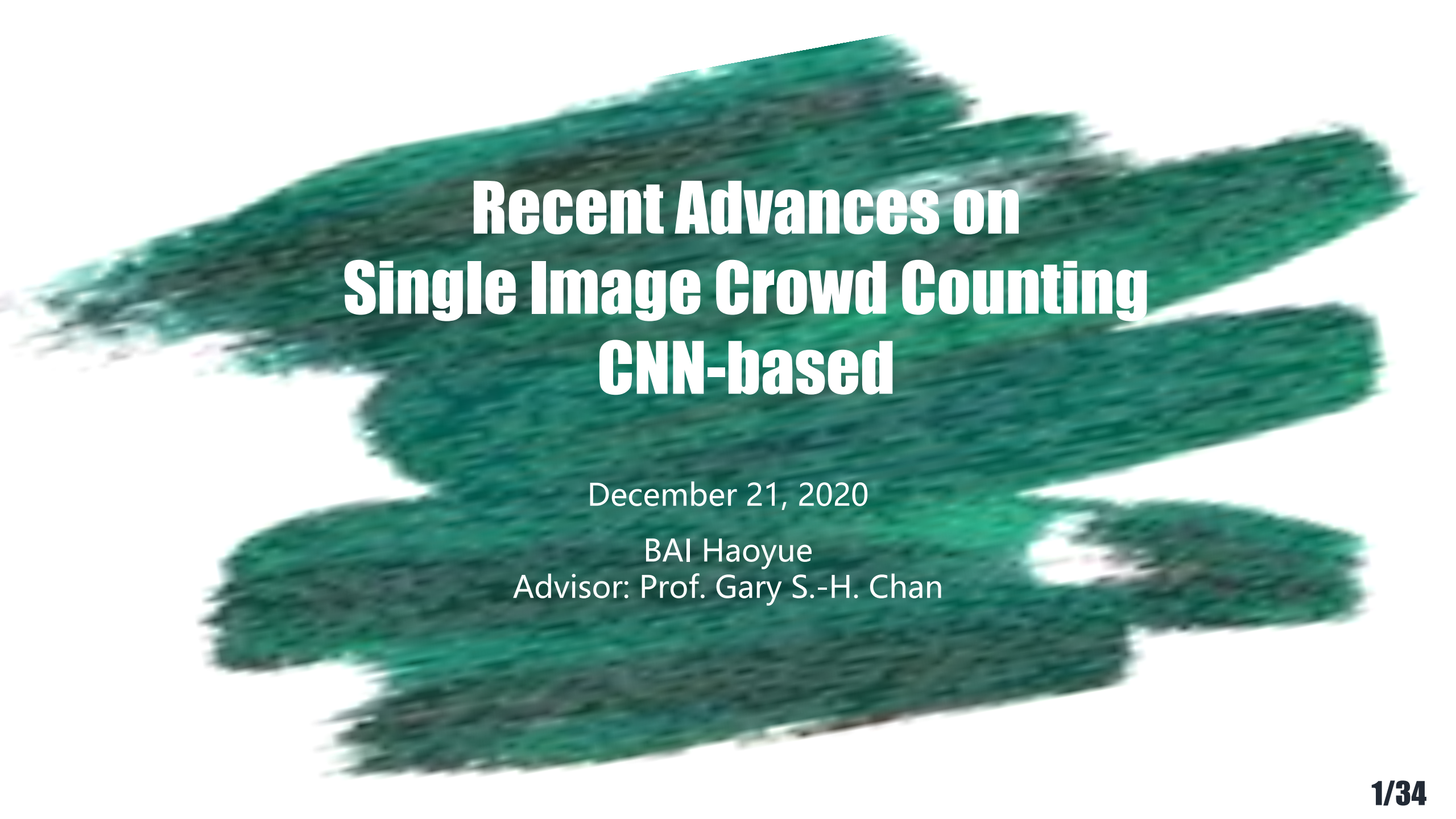


The background of the slide features a large, abstract, dark green brushstroke that sweeps across the frame from the top left towards the bottom right. The stroke has a textured, painterly quality with varying shades of green and some darker, almost black, areas within the main stroke.

Recent Advances on Single Image Crowd Counting CNN-based

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Outline

- 1 Introduction**
- 2 Datasets and Evaluation Metrics**
- 3 Deep Neural Network Designs**
- 4 Optimization Strategies**
- 5 Data Processing Methods**
- 6 Conclusion and Future Directions**



Introduction

1 Single Image Crowd Counting

- Single image **crowd counting** is an important class of computer vision techniques for a **wide range** of intelligent surveillance applications.
- This work covers the **design considerations** and **recent advances** with respect to single image crowd counting, and summaries **more than 100** approaches proposed since 2015.



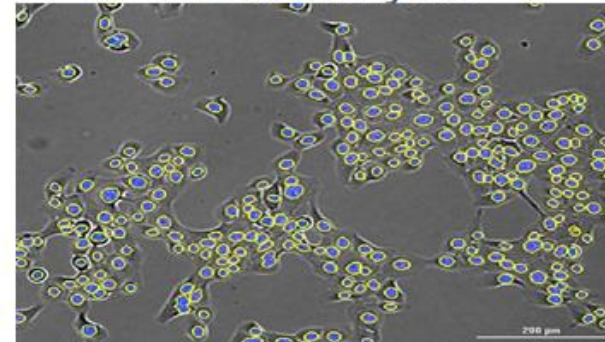
Public safety



Congestion avoidance



Flow analysis



Medical & Biological
image processing

1 Traditional approaches

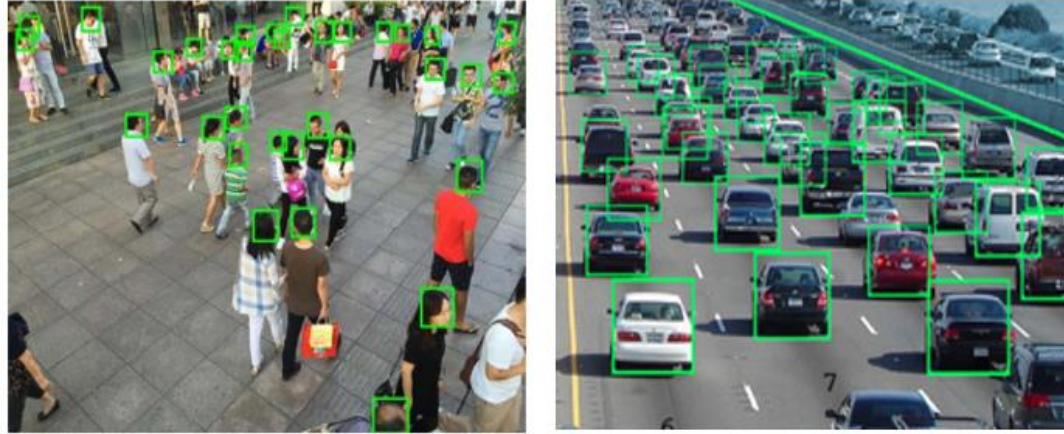


Figure 1: Traditional detection-based methods.
Faster R-CNN (CVPR 2016 Best Paper)

➤ Single Image Crowd Counting:

Crowd counting aims to estimate the number of object (e.g., people in this survey) in an image of **unconstrained** (free view) scenes.

➤ Traditional Approaches:

1. Counting by detection: detection of people based

Weakness: Accurate detect is difficult for highly crowded scenes.

2. Counting by regression: regression on people counts

Weakness: Less interpretable; lacks location information.

1 Crowd Counting via Density Map Regression

Table 1: Summary of traditional approaches with density map estimation methods.

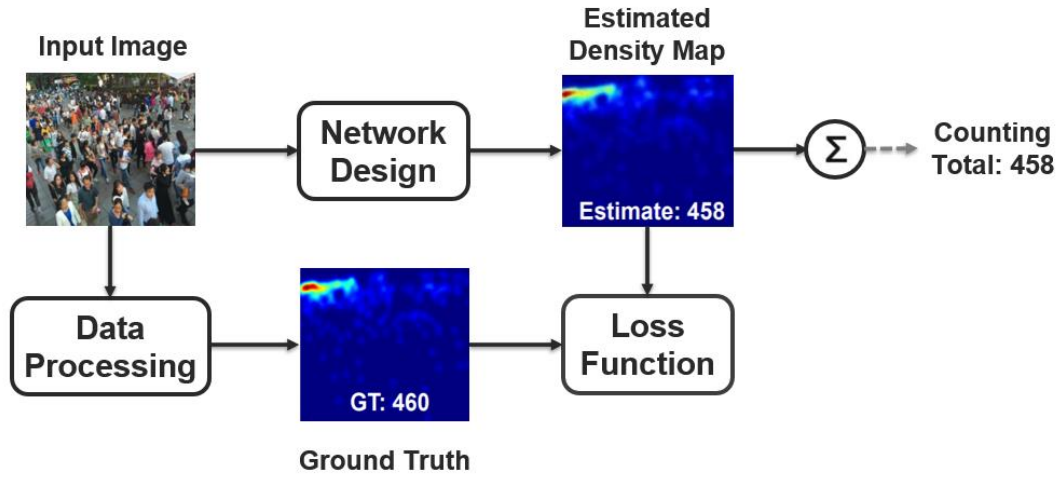
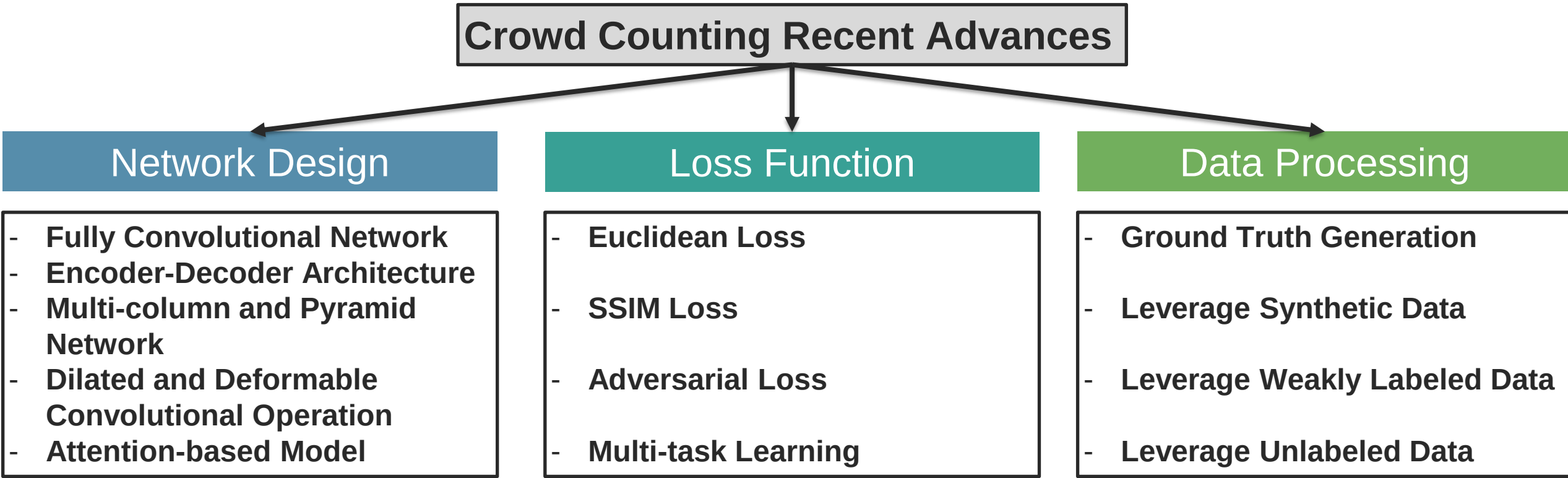


Figure 2: Basic crowd counting system flow.

Category	Attributes	Counting Accuracy	Locate Accuracy	Annotation Complexity	Limitations	Schemes
Detection-based approach	Detect then count; early methods	Low	High	High; object-level	Accurate detect is difficult for highly crowded scenes	Detect people or heads [37], [33]
Regression-based approach	Direct learn to regress the count	Medium	N/A	Low; image-level	Less interpretable; lacks location information	Regress the count [8], [7]
Counting via density map estimation	Robust than detection and regression based methods	High	Medium	Medium; point-level	Low counting accuracy in relatively low crowd scenes	MCNN [100], CSRNet [36], SANet [5]

- This survey mainly focus on recent advanced crowd counting techniques using CNN-based approaches via density map estimation.
- CNN-based density map estimation crowd counting methods: the input image capturing a crowd scenes is mapped to its corresponding density map. The ground truth density map indicates the number of people per pixel in the image, which is generated by reprocessing the original annotation.

Crowd Counting Recent Advances



- Recent advanced crowd counting techniques using CNN-based approaches via density map regression, can be roughly divided into three categories: network designs, loss functions and data processing manners for training data.

2

Datasets and Evaluation Metrics

2 Publicly Available Crowd Counting Datasets



Figure 4: Visualization of publicly available datasets for crowd counting.

- The most widely used crowd counting datasets to evaluate the performance of the crowd counting models:
 - Single image: NWPU, UCF-QNRF, ShanghaiTech A & B, UCF_CC_50.
 - Video sequence: FDST, WorldExpo'10, UCSD, Mall.
 - Synthetic: GCC.

2

Publicly Available Datasets Statistics

Table 1: Comparison of publicly available datasets statistics for crowd counting.

Dataset	Year	Average Resolution	Number of Images	Total	Min Count	Max Count	Average Count
NWPU-Crowd [76]	2020	2191 × 3209	5,109	2,133,375	0	20,033	418
UCF-QNRF [25]	2019	2013 × 2902	1,535	1,251,642	49	12,865	815
GCC [62]	2019	1080×1920	15,212	7,625,843	0	3,995	501
Fudan-ShanghaiTech [16]	2019	1080 × 1920	15,000	394,081	9	57	27
ShanghaiTech Part A [97]	2016	589 × 868	482	241,677	33	3,139	501
ShanghaiTech Part B [97]	2016	768 × 1024	716	88,488	9	578	124
WorldExpo'10 [91]	2015	576 × 720	3,980	199,923	1	253	50
UCF_CC_50 [24]	2013	2101 × 2888	50	63,974	94	4,543	1,280
Mall [9]	2012	240 × 320	2,000	62,325	13	53	31
UCSD [6]	2008	158 × 238	2,000	49,885	11	46	25

- Some considerations on how to choose crowd counting datasets:
 - Datasets with high resolutions usually show better image and visual quality.
 - Datasets with larger number of images cover more diverse scenes.
 - Datasets with more object number are more challenge and suitable for evaluating crowd counting algorithms.

Evaluation Metrics and design considerations

➤ **Accuracy: counting accuracy and location accuracy.**

- counting accuracy: need to consider large scale variations and isolated clusters problems (long-range contextual information). MAE and MSE are used to quantitatively evaluate counting accuracy.
- location accuracy, density map estimation methods preserves spatial information, while total number regression approaches lacks spatial information.

➤ **Quality of density maps: resolution and visual quality**

- PSNR and SSIM are used to quantitatively evaluate counting accuracy.

➤ **Complexity: computational complexity and annotation complexity**

- Number of parameters is used to evaluate computational complexity

➤ **Flexibility and robustness**

- Flexibility: model can process images of arbitrary size and can model different kinds of objects (e.g., non-rigid objects).
- Robustness: robustness to distribution shift, where test data come from another distribution (w.r.t. the training one).



Deep Neural Network Designs

3 Fully Convolutional Neural Networks

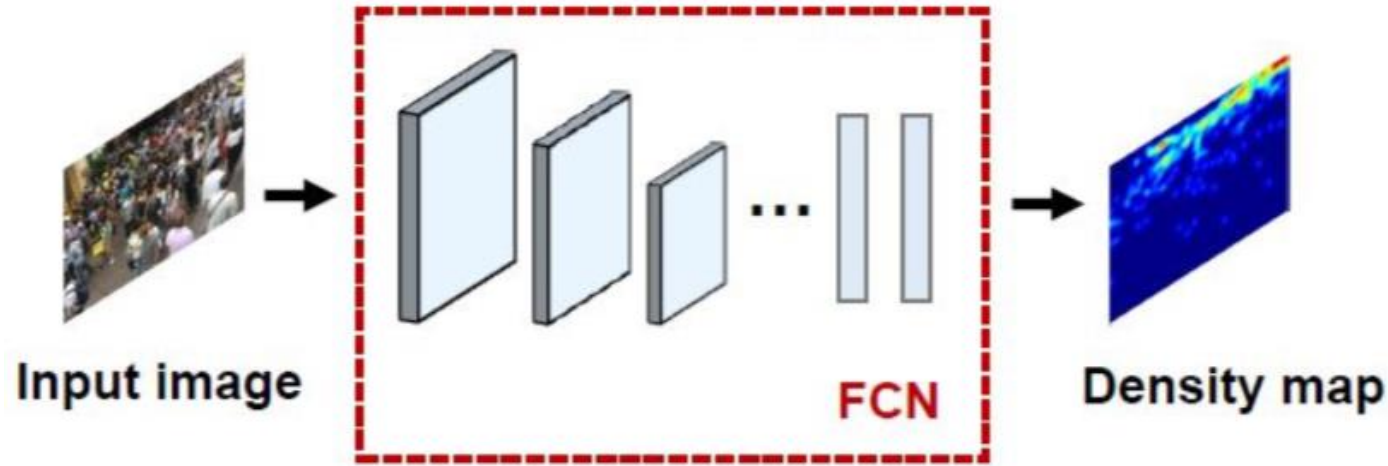


Figure 5: Illustration of fully convolutional networks for crowd counting.

- Fully convolutional neural networks (FCN) modified the existing CNN architectures and replace fully-connect layers with fully convolutional layers
- **Advantages:** FCN can analyze images of arbitrary size.
- **Limitations:**
 - Low resolution density maps
 - Accurate counting is difficult for images with large scale variations (two objects of same physical size would appear different size in an image).

3 Encoder-Decoder Architecture

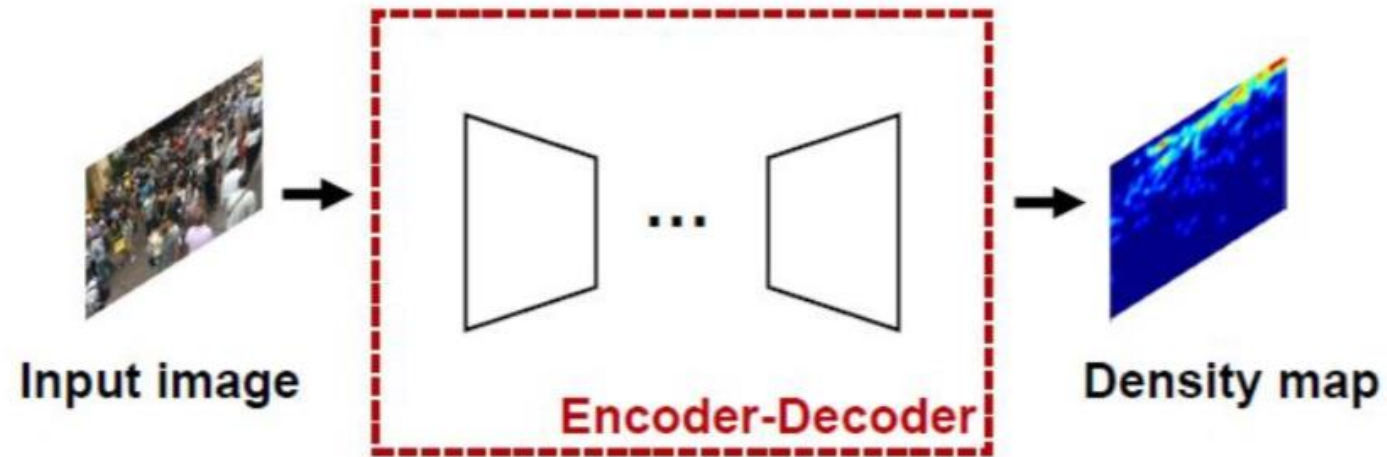


Figure 6: The architecture of encoder-decoder based model.

- Many CNN-based crowd counting approaches are in some kind of encoder-decoder form.
- **Advantages:** Generate high-resolution and high-quality density maps.
- **Limitations:**
 - Not consider scale variations problems
 - Not consider the local and global contextual information.

3 Multi-column Architecture

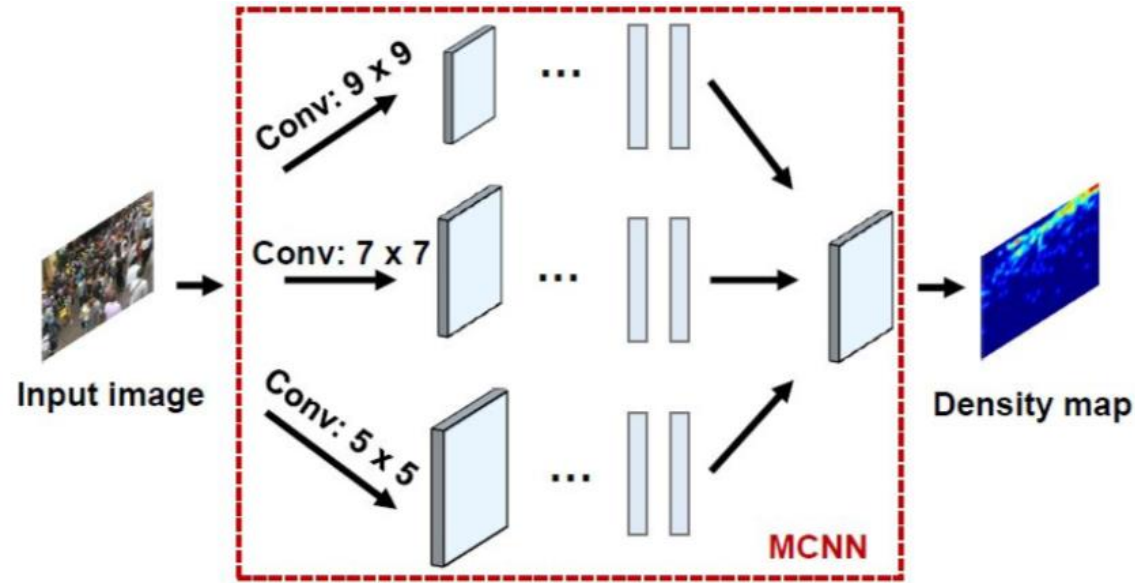


Figure 7: Representative approach that leverage the multi-column based architecture for crowd counting.

- Multi-column architecture for crowd counting incorporates multi-column architecture with different kernel sizes to extract different scale features and achieve accurate prediction.
- **Advantages:** schemes to address scale variations.
- **Limitations:** The scale diversity is restricted by the number of columns.

3 Pyramid Architecture

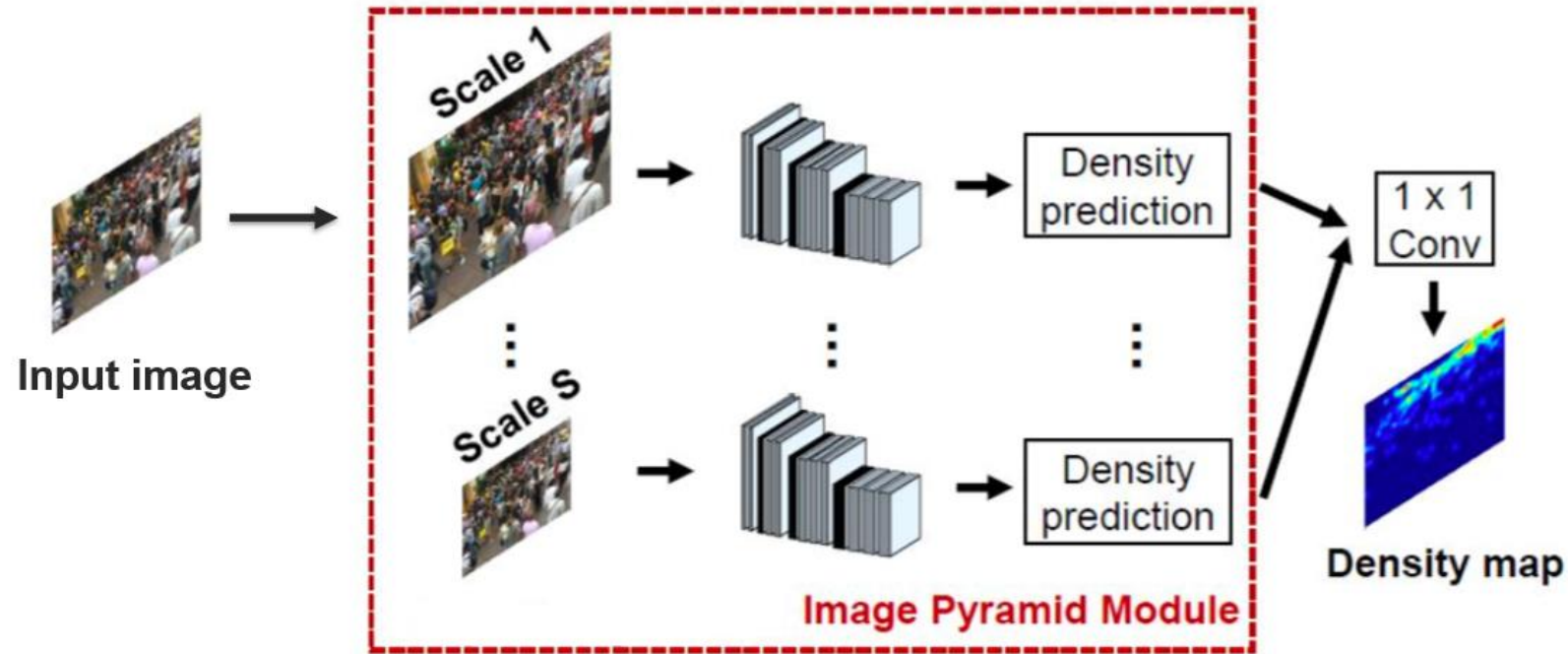


Figure 8: Illustration of image pyramid-based model for crowd counting.

- Pyramid architecture for crowd counting consists of two subgroups: image pyramid and feature pyramid pooling.
- **Advantages:** Another techniques to address scale variations.
- **Limitations:** High computational complexity.

3 Dilated and Deformable Convolutional Operations

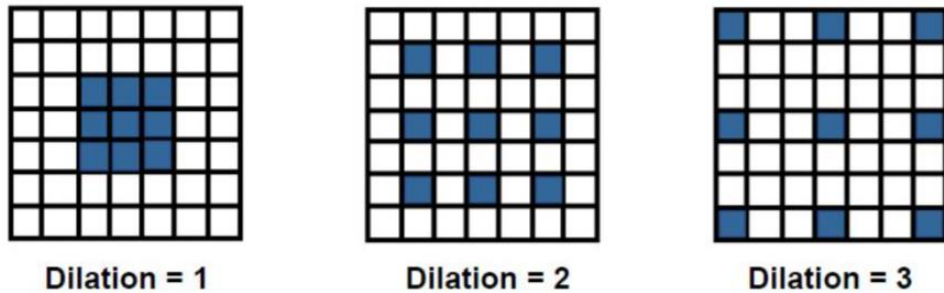


Figure 9: The details of the dilated convolutional operations.

- Deformable convolution is a kind of spatial sampling locations augmenting in the modules with additional offsets.
- **Advantages:** Model with additional offsets for non-rigid objects.
- **Limitations:** High computational complexity.

- Dilated convolution introduces the dilated rate (define a spacing between the weights of kernel) to the convolutional layers.
- **Advantages:** Enlarge receptive field without increasing the computational cost.
- **Limitations:** Not consider the multi-scale features.

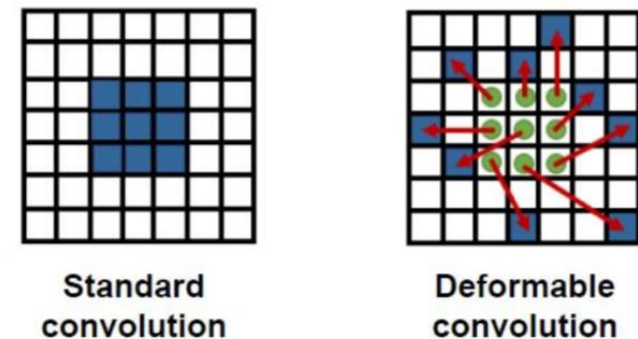


Figure 10: The details of the deformable convolutional operations.

3 Attention-based Model

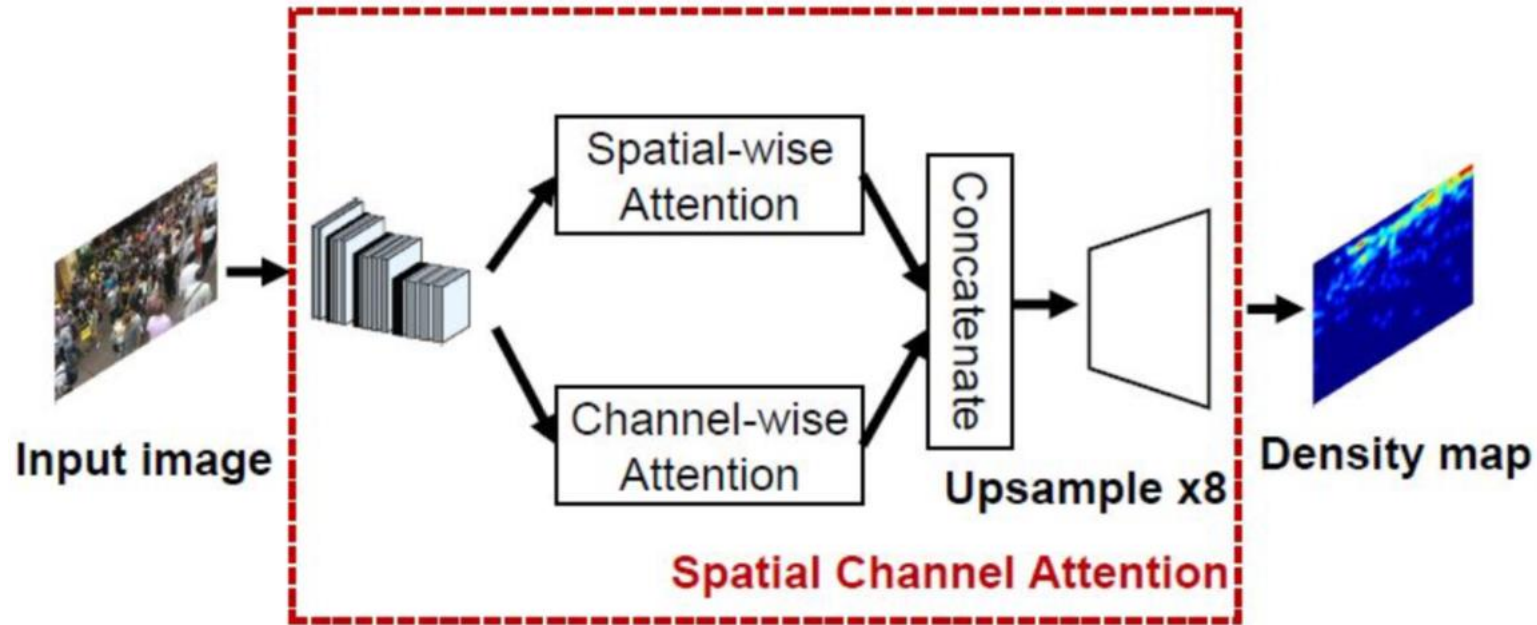


Figure 11: Illustration of attention scheme for crowd counting.

- Attention mechanisms can be roughly divided into two subgroups: hard attention and soft attention.
- **Advantages:** Account for both short-range and long-range interdependencies.
- **Limitations:** Rely on pixel-wise loss function.

3

Summary of Neural Network Designs

Table 3: Summary of network design considerations.

Category		Scheme	Advantages	Computational Complexity	Limitations
Fully convolution neural networks		FCN [50]	Can analyze images of arbitrary size	Low	Low-resolution density maps
Encoder-Decoder Architecture		SANet [5]	Able to generate high-resolution density maps	Low (0.91M)	Not consider scale variation
Multi scale features	Multi-column architecture	MCNN [100]	Extract multi-scale features with multi-column architecture	Low (0.13M)	The scale diversity is restricted by the number of columns
	Pyramid architecture	CP-CNN [68]	Extract multi-scale features with pyramid architecture	High (68.4M)	High computational complexity
Advanced convolution operations	Dilated convolution operations	CSRNet [36]	Enlarge receptive field without increase the computational cost	Medium (16.26M)	Not consider the non-rigid objects
	Deformable convolution operations	ADCrowdNet [43]	Learnable additional offsets for better modeling non-rigid objects	High	High computational complexity
Attention-based Model		SCAR [18]	Capture local and global contextual information	Medium	Rely on pixel-wise loss function

3

Summary of Neural Network Designs

Table 4: Comparisons of network design considerations.

Typical Schemes			ST PartA				ST PartB		UCF_CC_50		UCSD	
Methods	Year	Column	MAE	MSE	PSNR	SSIM	MAE	MSE	MAE	MSE	MAE	MSE
FCN [50]	2016	Single	126.5	173.5	N/A	N/A	23.8	33.1	338.6	424.5	N/A	N/A
MCNN [100]	2016	Multi	110.2	173.2	21.40	0.52	26.4	41.3	377.6	509.1	1.07	1.35
CP-CNN [68]	2017	Multi	73.6	106.4	21.72	0.72	20.1	30.1	295.8	320.9	N/A	N/A
SANet [5]	2018	Single	67.0	104.5	N/A	N/A	8.4	13.6	258.4	334.9	1.02	1.29
CSRNet [36]	2018	Single	68.2	115.0	23.79	0.76	10.6	16.0	266.1	397.5	1.16	1.47
ADCrowd [43]	2019	Single	63.2	98.9	24.48	0.88	8.2	15.7	266.4	358.0	1.10	1.42
SCAR [18]	2019	Double	66.3	114.1	23.93	0.81	9.5	15.2	259.0	374.0	N/A	N/A
HyGnn [47]	2020	Double	60.2	94.5	N/A	N/A	7.5	12.7	184.4	270.1	N/A	N/A

- Compare the typical schemes discussed above on the publicly available crowd counting datasets:
 - SANet shows better performance, compared with FCN. The generate density map of FCN is only $\frac{1}{4} \times \frac{1}{4}$ of original input image, while SANet is able to generate high-resolution density maps.
 - MCNN and CP-CNN considers scale variations and shows better performance than FCN. CP-CNN achieves better counting accuracy and visual quality than MCNN, while the number of parameters of CP-CNN is much larger than MCNN.
 - CSRNet and ADCrowd achieve better counting accuracy and visual quality than MCNN and CP-CNN, shows effectiveness of advanced convolutional operations.



Optimization Strategies

4 Summary of Optimization Strategies

Table 5: Summary of some recent advanced loss functions for crowd counting.

Category	Advantages	Limitations	Schemes
Euclidean loss	Flexible; widely used	Not consider context information and visual quality	MCNN [100]
SSIM loss	Variants of structural similarity loss to learn local correlation	Hard to learn the local correlation with various scales	DSSINet [41]
Adversarial loss	Generate high-quality crowd density maps	Degrade counting accuracy	ACSCP [48]
Multi-task learning	Fuse Euclidean loss with direct global count constrain	Sensitive to hyper-parameters	HA-CCN [70]
Others	Efficient divide and conquer manner	Computational expensive	S-DCNet [89]

4 Comparison of Optimization Strategies

Table 6: Quantitative comparison of state-of-the-art crowd counting approaches.

Scheme	Year	Multi scale	Dilated	Deform	Atten	Loss function	ST-A MAE	ST-A MSE	ST-B MAE	ST-B MSE
CSRNet [36]	2018		√			Euclidean loss	68.2	115.0	10.6	16.0
ADCrowd [43]	2019			√	√	Euclidean loss	63.2	98.9	8.2	15.7
DSSINet [41]	2019	√	√			SSIM Loss	60.63	96.04	6.8	10.3
ACSCP [48]	2019	√	√			Adversarial loss	85.2	137.1	15.4	23.1
S-DCNet [89]	2019	√				Divide-conquer	58.3	95.0	6.7	10.7
HA-CCN [70]	2020				√	MSE loss with global counting	62.9	94.9	8.1	13.4

- In the field of crowd counting, the loss functions are used to measure the gap of predicted density map and the ground truth, and optimize the model.
- CSRNet and ADCrowd based on Euclidean loss but with different networks designs and shows different counting accuracy, which shows L2 loss is flexible and widely used in early approaches.
- DSSINet achieves better counting accuracy than ACSCP with similar network design considerations. The poor performance of ACSCP may probably due to the adversarial loss
- HA-CNN shows better performance than ADCrowdNet even without deformable convolutional operations , which shows the effectiveness of multi-task learning with global counting constrain.
- S-DCNet also achieves satisfactory counting accuracy, which confirms the effectiveness of divide and conquer manner but is computational expensive.



Data Processing Methods

5 Ground Truth Generation

- For the ground truth generation, geometry-adaptive kernels are widely used to achieve accurate estimation in the highly congested scenes: $F(x) = \sum_{i=1}^N \delta(x - x_i) \times G_{\sigma_i}(x)$, with $\sigma_i = \beta \overline{d_i}$
- **Advantages:** More accurate than fixed kernel ground truth generation methods.
- **Limitations:** Sensitive to hyper-parameters. And not robust to non-rigid objects.

5 Leverage Synthetic Data for Crowd Counting



- In the field of crowd counting, synthetic data can be used to tackle the over-fitting problem for crowd counting.
- **Advantages:** Reduce labeling effort; enhance accuracy; improve robustness.
- **Limitations:** Large domain gap from synthetic to real data.



5 Leverage Weakly Labeled Data

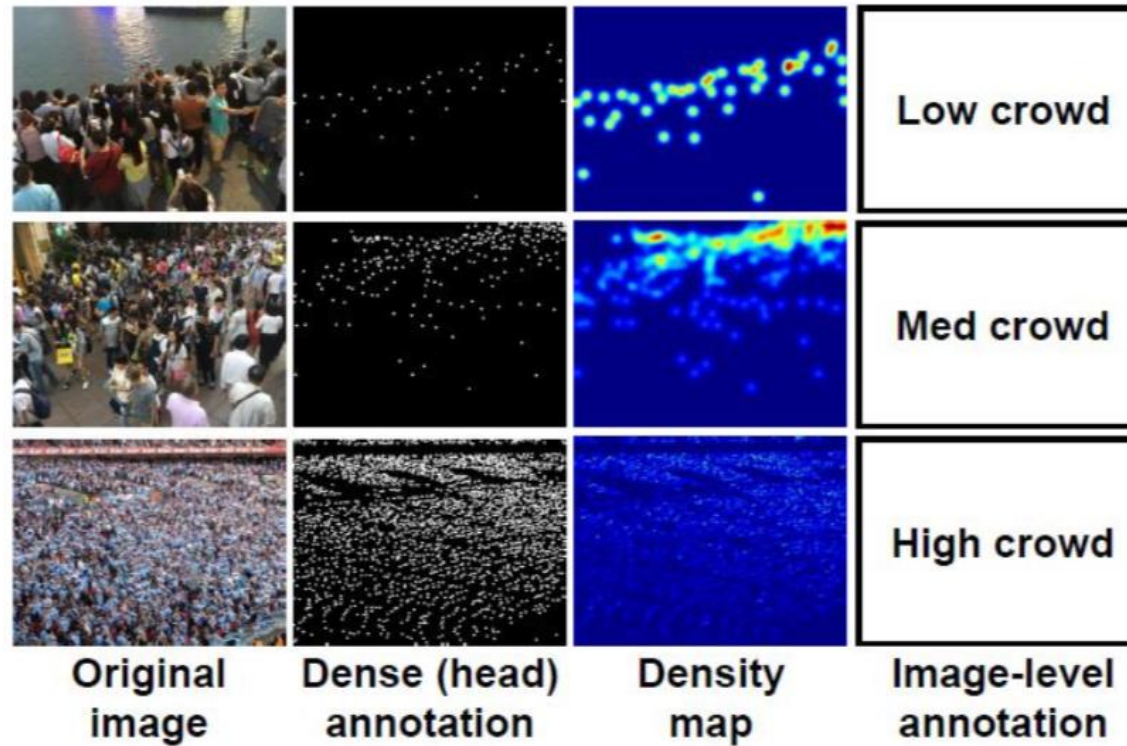


Figure 13: A weakly supervised learning setup for crowd counting.

- Leverage image-level labels instead of the labor-intensive point-wise annotation process.
- **Advantages:** Low annotation complexity.
- **Limitations:** Still requires annotations; requires more task specific knowledge.

5 Leverage Unlabeled Data

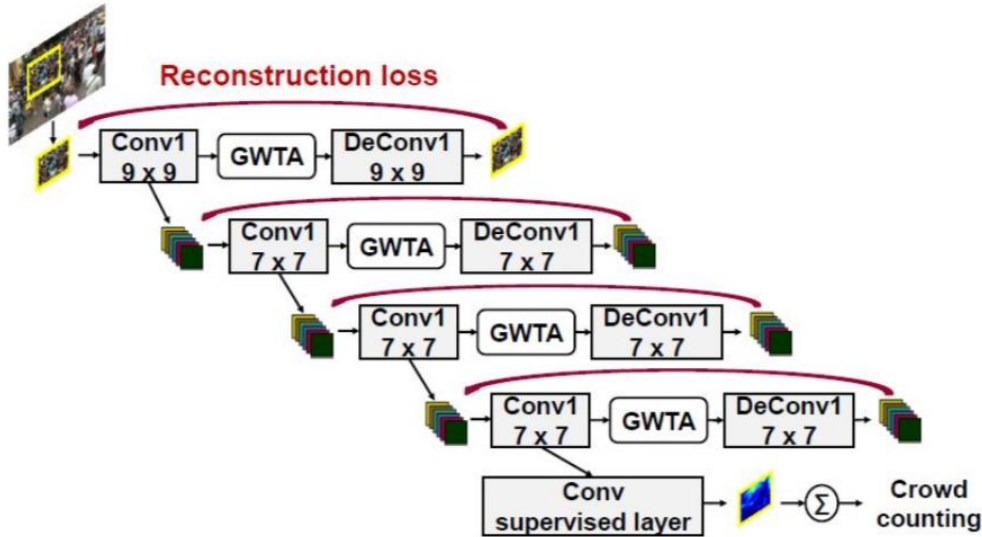


Figure 14: The architecture of the almost unsupervised learning framework for crowd counting.

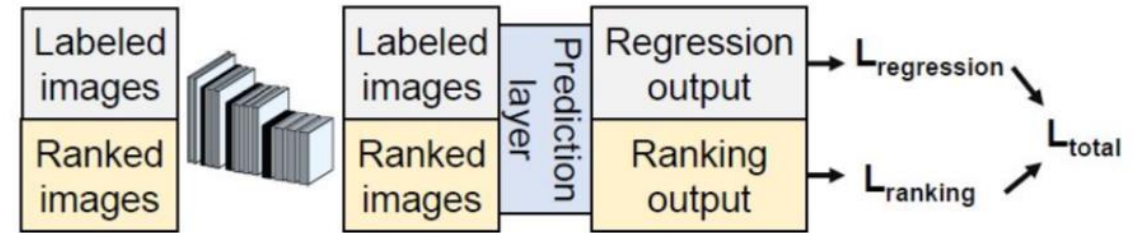


Figure 15: A self-supervised learning setup for crowd counting.

- Most of the parameters are trained on unlabeled data using reconstruction loss or in a self-supervised learning manner.
- **Advantages:** Unlabeled data is abundantly available and cheap.
- **Disadvantages:** Larger amount of data requires more computational resources and training time.

5 Summary of Data Processing Methods

Table 7: Summary of recent advanced data processing manners for crowd counting.

Category	Schemes	Advantages	Limitations
Ground truth generation	MCNN [100]	Adaptive Gaussian kernel to accommodate different scales	Not flexible to non-rigid object
	BL [49]	Bayesian loss to model non-rigid objects	More reliable supervision but suffers in large varied scales
Leverage synthetic data	CCWld [80]	Reduce labeling effort; enhance accuracy; improve robustness	Large domain gap from synthetic to real data
Leverage weakly labeled data	HA-CCN [70]	Low annotation complexity	Still requires weakly annotations and task specific knowledge
Leverage unlabeled data	L2R [45]	Low annotation cost; abundantly available	Large amount of data requires more training time

5 Comparison of Data Processing Methods

Table 8: Analysis of state-of-the-arts crowd counting approaches with different data processing manners.

Typical Schemes			ST PartA		ST PartB		UCF_CC_50		UCF-QNRF	
Methods	Year	Columns	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
MCNN [100]	2016	Multi	110.2	173.2	26.4	41.3	377.6	509.1	277	426
L2R [45] (Query by example)	2018	Double	72.0	106.6	14.4	23.8	291.5	397.6	N/A	N/A
L2R [45] (Query by keyword)	2018	Double	73.6	112.0	13.7	21.4	279.6	388.9	N/A	N/A
BL [49]	2019	Single	88.7	154.8	62.8	101.8	229.3	308.2	88.7	154.8
CCWld [80]	2019	Single	64.8	107.5	7.6	13.0	N/A	N/A	102.0	171.4
HA-CCN [70]	2020	Single	58.3	95.0	6.7	10.7	256.2	348.4	118.1	180.4

- BL achieves better performance compared with MCNN. The good performance of BL may due to the Bayesian loss used to better model the non-rigid objects.
- CCWld shows much better counting accuracy than MCNN. This shows the CCWld enhances the performance of crowd counting and also improves the robustness.
- The performance of HA-CNN is much better than other state-of-the-arts. After carefully designing the networks and loss functions, weakly supervised crowd counting achieves much better accuracy with relatively low annotation complexity.
- The MAE and MSE of L2R is lower than MCNN and BL. This confirms that leveraging the abundantly available unlabeled data improves counting performance.



Conclusion and Future Directions

6 Future Directions

➤ Conclusion:

- This work covers the design considerations and recent advances with respect to single image crowd counting, and summaries more than 100 algorithms since 2015. The research field of crowd counting still needs further study.

➤ Future directions:

- Efficient network and CNN feature design
- Weakly supervised and unsupervised crowd counting
- Crowd counting in videos
- Multi-view fusion for crowd counting

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THANKS